CH3 Mathematical Foundations of Symbolic Aesthetic Tension

This chapter formalizes the internal mathematical representation that allows an AI model to perceive symbolic music without audio. Music is defined as a sequence of discrete events. Each event contains pitch, duration, metric position, harmonic function, and symbolic context.

Define tension tau as a scalar assigned to each musical event. Tension is determined by harmonic gravity, melodic interval potential, and rhythmic momentum.

Harmonic gravity is defined as the distance between the harmonic embedding of an event and the tonal center. A small distance means stability, a large distance means tension.

Melodic interval potential is computed from the size and direction of the interval between successive pitches. Large intervals create higher potential. Upward intervals create expectation. Downward intervals create resolution.

Rhythmic momentum is the combined weight of duration and metric position. Shorter durations increase tension. Weak beats increase tension. Strong beats decrease tension.

The total tension of an event is defined as the sum of harmonic gravity, interval potential, and rhythmic momentum.

The sequence of tensions produces a tension curve across the entire score. This curve is the model’s internal equivalent of musical "feeling" or "pressure."

Resolution probability is defined as the likelihood that tension will decrease within the next few events. High resolution probability indicates expected closure. Low probability indicates continuation.

Aesthetic latent space is a multidimensional geometric space into which each event is embedded. Every embedding contains tension, gravity, interval potential, rhythmic weight, and symbolic features. This space allows the model to compare patterns, predict structure, and generate internal analogies without needing audio.

This chapter establishes the mathematical foundation for symbolic auditory cognition inside an AI system. It demonstrates that music can be understood as structural tension rather than waveform phenomena, enabling training without audio inputs.